

GTU CSE 241 / 501 Quiz

Std Code: 210

Std Number:

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using namespace std;

class Person{

public:

Person(): name(""), age(0){} // default +

Person(const string& nm, int do+o): name(nm), age
(do+o){if (age < 0) exit(1);}

void set-name(const string& do+o){ name = do+o;}

string get-name() const { return name;}

void set-age(int do+o){ age = do+o;}

int get-age() const { return age;}

Person(const Person& p): name(p.name), age(p.age)
{ // copy constructor }

friend ostream& operator<<(ostream& os, const Person& p){ os<<p.name<<" "<<p.age<<endl; return os;}

protected: // to occur in derived class
string name;

int age; // checked for negative numbers

};

```

class Student : public Person {
public:
    Student() : Person(), GPA(0.0) {}
    Student(const string& nom, int oa, double gp) : Person(
        nom, oa), GPA(gp) { if (GPA < 0.0) exit(1); }
    Student(double g) : Person(), GPA(g) { if (g < 0) exit(1); }
    Student(const Student& st) : Person(st), GPA(st.GPA)
    {}

    void setGPA(double gp) { if (gp < 0.0) exit(1);
                           GPA = gp; }

    double getGPA() const { return GPA; }

    friend ostream& operator<<(ostream& os, const Student& st) {
        os<<getNome()<<" is "<<getAge()<<" years old
        and he has "<<GPA<<endl; return os; }

    // we could directly used nome and age.

```

private:

double GPA; // checked invalid gpa

};